# Installing Docker on Your Machine

In this train, we'll guide you through the process of installing Docker on your local machine. We'll ensure that you have all the required software to begin Docker-based development.

## Learning Objectives

By the end of this train you should be able to:

- Successfully install required software and prepare your machine for Docker-based development.
- Understand how to configure Docker for different operating systems.
- Learn how to verify Docker installation and ensure it is working correctly.

## How to Install Docker on a Windows Computer

Follow these steps to install Docker on a Windows machine using the terminal:

Open PowerShell as Administrator and enable Windows Subsystem for Linux (WSL 2):

```
wsl --install
```

Restart your computer and set WSL to use version 2:

```
wsl --set-default-version 2
```

Install the required dependencies:

```
wsl --update
```

Download and install Docker from the official repository:

```
Invoke-WebRequest -Uri https://get.docker.com -UseBasicParsing | Invoke-
Expression
```

Verify Docker installation:

```
Verify Docker installation:
```

Start the Docker service:

```
sudo service docker start
```

Test Docker by running:

```
docker run hello-world
```

## How to Install Docker on Mac Os

Follow these steps to install Docker on Mac using the terminal:

Open the terminal and install Homebrew if not already installed:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Install Docker:

```
brew install docker
```

Verify the installation:

```
docker --version
```

Start Docker:

```
dockerd &
```

` Test Docker with:

```
docker run hello-world
```

## How to Install Docker on a Linux Machine (ubuntu Os)

Follow these steps to install Docker on an Ubuntu Linux machine using the terminal:

Update your package manager:

```
sudo apt update && sudo apt upgrade -y
```

Install required dependencies:

```
sudo apt install -y apt-transport-https ca-certificates curl software-
properties-common
```

Add the Docker GPG key:

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/docker-archive-keyring.gpg
```

Add the Docker repository:

```
echo "deb [arch=amd64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
```

Update package lists again:

```
sudo apt update
```

Install Docker:

```
sudo apt install -y docker-ce docker-ce-cli containerd.io
```

Start and enable Docker:

```
sudo systemctl start docker
sudo systemctl enable docker
```

Add your user to the Docker group:

```
sudo usermod -aG docker $USER
```

Verify installation:

```
docker --version
```

Test Docker with:

```
docker run hello-world
```

## Conclusion

Docker is now installed on your machine using only the terminal! You can start using it to create, manage, and run containers. Ensure that Docker is running correctly by testing it with the hello-world image. If you run into any issues, refer to the official Docker documentation for troubleshooting.

**Key Takeaways** ✖

- **Takeaway 1:** Terminal-based installation - This guide shows how to install Docker without a graphical interface.
- **Takeaway 2:** Platform-specific steps - The installation differs for Windows, Mac, and Linux.
- **Takeaway 3:** Verification is crucial - Always confirm installation by checking the version and running a test container.
- **Takeaway 4:** Linux users need extra setup - Adding the user to the Docker group prevents permission errors.

- **Takeaway 5:** Start using Docker - Now that Docker is installed, you can begin working with containers!